

ARM® Cortex™-M3 Processor

Delivering Cost-Sensitive High-Performance
32-Bit Devices

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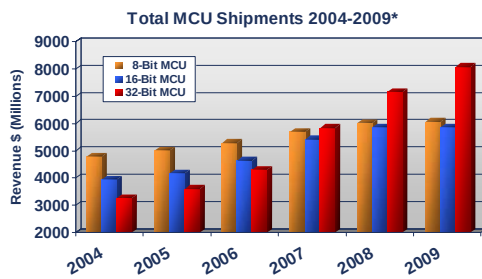


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Deeply Embedded Processor Market



*Semico Research: Embedded Control 2005 (Q2 2005)

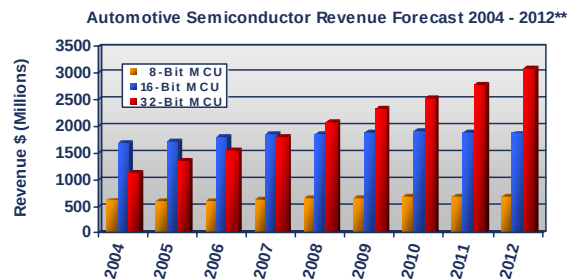


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Deeply Embedded Processor Market



**Strategy Analytics Automotive Semiconductor Demand
Forecast 2003-2012 (Sept 2005)

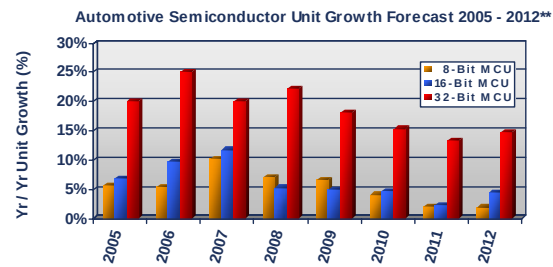


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Deeply Embedded Processor Market



**Strategy Analytics Automotive Semiconductor Demand Forecast 2003-2012 (Sept 2005)



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Industrial Evolution

- What is driving the migration from 8- and 16-bit?
 - Increasing complexity
 - Features: Additional features and increasingly complex algorithms
 - Connectivity: Increasing integration with external systems
 - Reducing costs
 - Pricing: 32-bit devices becoming available below \$3
 - Aggregation: Multiple 8/16-bit devices replaced by single 32-bit device
 - Code density: Improving code density reduces memory size
 - Accelerating time to market
 - Abstraction: Rapid development utilizing high level tools
 - Platforms: Development of multiple products based on single platform

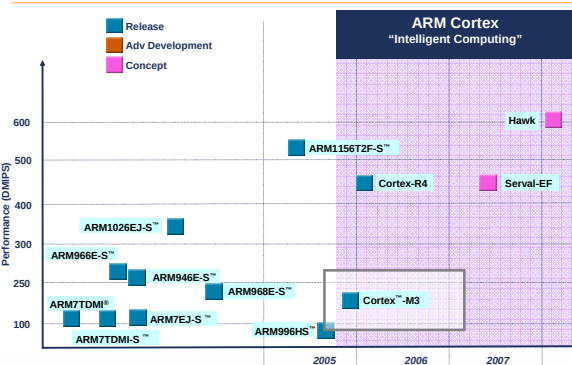


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Embedded Processor Roadmap



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Cortex-M3 Processor Benefits

- Delivery of low-power cost-sensitive 32-bit devices with simplified development and reduced time to market
- Ease of use
 - No assembler required
 - Advanced debug and trace
- High performance
 - Outstanding efficiency of 1.25 DMIPS/MHz
- Low power
 - Low-power design
 - Integrated sleep modes
- Lower cost devices
 - 30%+ reduction in code memory with performance of ARM 32-bit code
 - Smallest ARM processor



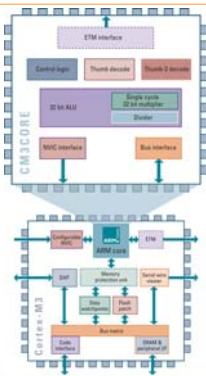
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Introduction to Cortex-M3 Processor

- Hierarchical processor integrating core and advanced system peripherals
- Cortex-M3 core
 - Harvard architecture
 - 3-stage pipeline w. limited branch prediction
 - Thumb®-2 and traditional Thumb only
 - ALU w. H/W divide and single cycle multiply
- Cortex-M3 Processor
 - Cortex-M3 core
 - Configurable interrupt controller
 - Bus matrix
 - Advanced debug components
 - Optional MPU & ETM



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Cortex-M3 Processor Performance

Cortex-M3 Processor Characterization (0.13um Metro)

Dhrystone Performance	1.25 DMIPS/MHz
Maximum Frequency	Approx 150MHz (Worst Case)
Power Consumption	CM3Core 0.06mW/MHz Cortex-M3 0.09mW/MHz
Area	CM3Core 0.14mm ² Cortex-M3 0.30mm ²

(Minimal Implementation @ Nominal 50MHz)

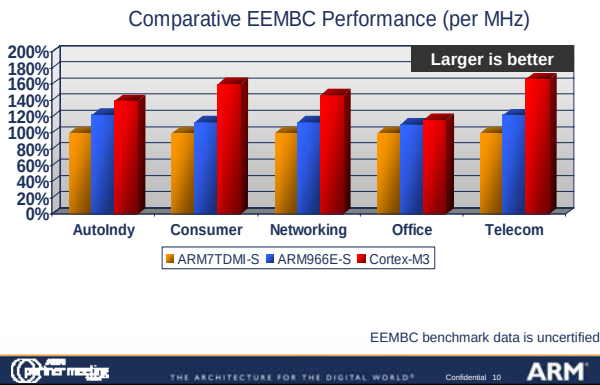


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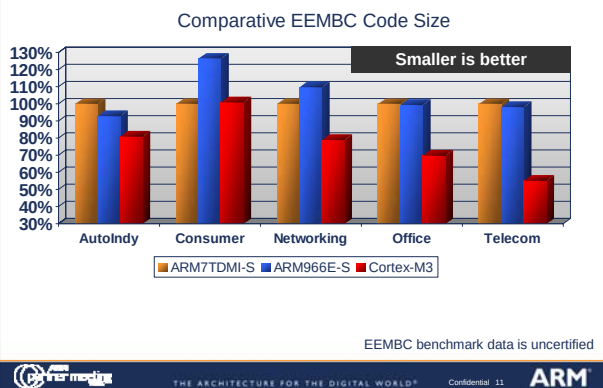
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Cortex-M3 Processor Performance



Cortex-M3 Processor Performance



Cortex-M3 Processor Technologies

- The Cortex-M3 processor enables designers to deliver high-performance deterministic devices with low dynamic power and reduced costs
 - Ease of use
 - Configurable interrupt controller
 - Integrated sleep modes
 - Hardware division and single cycle multiply
 - Atomic bit manipulation
 - Thumb-2 ISA
 - Integrated debug components

Ease-of-Use Lifecycle

Create

- Configurable NVIC
- Optional MPU and Embedded Trace Macrocell™ (ETM)
- Integrated debug
- Defined memory map

Program

- Simplified programmers model
- No assembly code required
- No interworking
- Powerful Thumb-2 ISA
- Hardware division

Debug

- 2-pin serial wire debug
- Interrupt aware
- 8 hardware breakpoints
- Small ETM



Intelligent Processors by ARM

Maintain

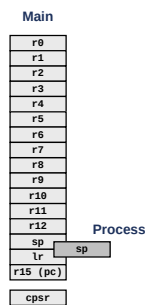
- Simplified code
 - No assembler or interworking
- Unified tools chain

Support

- Low-cost tools chain
- Profiling and event logging
- Flash patch program patching

Simplified Register Set

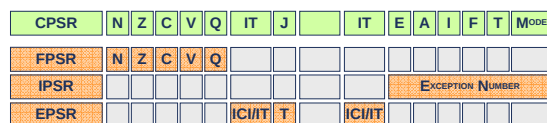
- Very simple, linear 4GByte address space
 - No data pages
 - No code pages
- Very compiler friendly
 - Flexible register scheme
 - Single-cycle multiply possible between any of the registers
 - Any register can be used as a pointer to data structures/arrays
- Same 'visible' register file as any other ARM architecture device
 - Allowing unified assembler to mimic equivalent ARM instructions in Thumb-2 mode



Program Status Registers

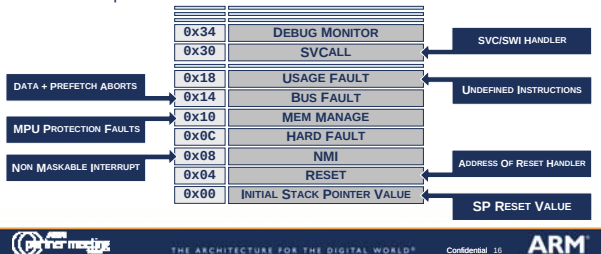
- ARMv7M architecture conceptually partitions the ARMv6 CPSR
 - FPSR – Flags are identical to previous ARMv6 processors
 - IPSR – Replaces mode and interrupt bits with current exception number
 - EPSR – Extends Thumb-2 if-then-else bits with interrupt continuation
 - Interrupted LDM/STM may resume using ICI rather than restarting
 - New MSR, MRS and CPS instructions allow access
 - For example:


```
MSR FPSR, r0
MRS r1, IPSR
```



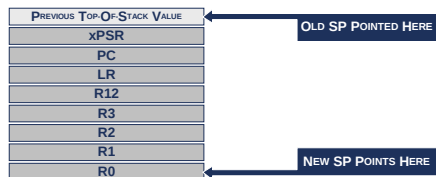
Vector Table

- ARMv7M architecture implements a re-locatable vector table
 - Contains the address of the function to execute for a particular handler
 - The address is fetched via the instruction port allowing register stacking to take place in parallel with lookup
 - The first sixteen entries are special with the others mapping to specific interrupts



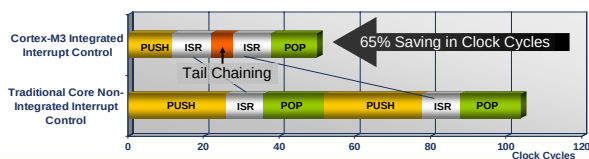
Exception Model

- Interrupts, synchronous faults and SVC exceptions
 - Exceptions cause current machine state to be stacked
 - Stacked registers conform to EABI
 - Exception handlers may simply be written in 'C' as:
`void IRQ(void) { /* my handler */ }`
 - Program status registers are merged together as xPSR



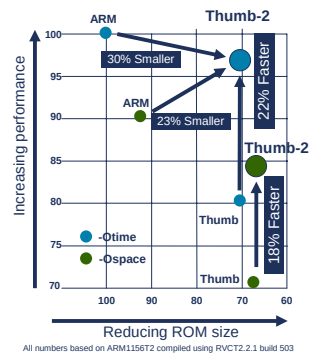
Advanced Interrupt Control

- The Cortex-M3 processor integrates an advanced nested vector interrupt controller (NVIC)
 - Easily configured for specific applications
 - 1-244 Interrupts and 1-255 priorities
 - Supports advanced features for next generation real-time applications
 - Tail-chaining of pending interrupts
 - Late-arrival interrupt handling and priority boosting / inversion



Thumb-2 Benchmarking

- Improves code size when starting from ARM code
 - Code size based on real world application data
 - 46 benchmarks
 - 48 applications
 - 9 Mbytes ROM
- Improves performance when starting from Thumb code
 - Performance numbers based on uncertified EEMBC numbers

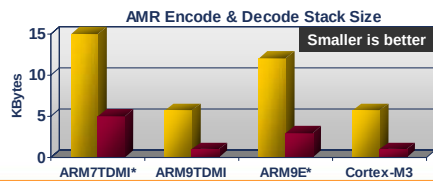
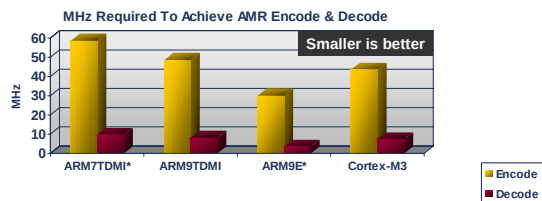


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Real World Application - AMR



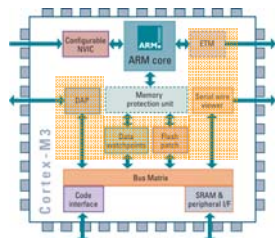
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Extensive Debug

- Core halt, reset and step
- Exposure of interrupt related information in the NVIC
- Profiling support
- Flash patch / breakpoints
- Data watchpoint and trigger (DWT)
- Complete trace support
 - Instruction only ETM
 - Data trace supported via DWT and HTM port



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Cortex-M3 Processor Everywhere?

The Cortex-M3 processor is ideal for a wide range of developers

- Migration from 8-bit and 16-bit domain
 - The Cortex-M3 processor offers simplicity of programming, exceptional performance and low power
 - Delivers a new "low entry point" to the ARM architecture
- Current 32-bit developers
 - Delivers higher performance at lower power and with less code
 - Enables either development of cheaper devices or enhanced applications within the boundaries of current devices
 - The Cortex-M3 processor is not just for MCUs – many uses already in complex ASICs
- The Cortex-M3 processor is the natural successor to the ARM7TDMI processor



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Performance Application Comparison

- Comparing processors is always challenging...
- Example: wireless system requiring 93 Dhrystone performance with 200KByte code size

	ARM7TDMI-S	Cortex-M3
Maximum Dhrystone Performance	0.93 DMIPS/MHz (ARM)	1.25 DMIPS/MHz
Frequency Required to Achieve 93 Dhrystone	100MHz	75MHz ← 25% Better
Power Consumed	39mW (100MHz x 0.39mW/MHz)	13.7mW (75MHz x 0.19mW/MHz) ← 64% Lower
Instruction Memory Required	200KB	130KB ← 35% Lower
Core Size (0.18um G Sage-X TSMC)	0.62mm²	0.43mm² (CM3Core) ← 30% Smaller



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Cost-Sensitive Applications

- Example MCU system requiring 15 Dhrystone performance with 20KByte code size

	ARM7TDMI-S	Cortex-M3
Dhrystone Performance	0.74 DMIPS/MHz (Thumb)	1.25 DMIPS/MHz ← 70% Better
Frequency required to achieve 15 Dhrystone	20MHz	12MHz ← 40% Better
Power Consumed	7.8mW (20MHz x 0.39mW/MHz)	2.3mW (12MHz x 0.19mW/MHz) ← 70% Lower
Instruction Memory Required	20KB	20KB
Core Size (0.18um G Sage-X TSMC)	0.62mm²	0.43mm² (CM3Core) ← 35% Lower
MCU Specific Features	N/A	No Assembler Required. Improved Interrupt Handling Atomic Bit Manipulation 2-pin Debug & Profiling Ability



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Migrating Applications

- Migrating applications to the Cortex-M3 processor can be as simple as a recompile
 - Tools also enable porting of optimized assembler code
- If source code is available, a simple recompile is all that is required
 - ARM C/C++ sources ~ **simple recompile**
 - Thumb C/C++ sources ~ **use as is** or simple recompile
 - ARM assembler sources ~ re-assemble using unified assembler
 - Thumb assembler sources ~ **use as is** or convert to Thumb-2
- ISRs may require simple editing to remove assembler stubs
 - No assembler stubs required due to hardware interrupt handling
 - Vector table points directly to compiled C routine
- Removal of boot code and system configuration assembler
 - No boot routine is needed
 - Hardware loads the stack pointer and initial PC from memory and enters as a normal C function



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Broad Tools Ecosystem

- RealView® DEVELOP family of tools
 - RealView Development Suite 3.0
- RealView CREATE family of tools
 - RealView SoC Designer
- RealView Microcontroller Developer Kit



- Third-party compiler and debug support
 - IAR, Green Hills, Lauterbach



- Open source support

- Code Sourcery



- Released RTOS support

- Express Logic, Mentor Graphics

- AutoSAR, eCOS, uC-OS II, FreeRTOS



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Automotive and Industrial Applications

- General purpose MCU
- Data acquisition
- White goods
- Dash
- Body control
- Breaking systems
- Engine control



The Cortex-M3 processor delivers:

- High performance
- Fast interrupt handling
- Lower cost 32-bit devices
- Low power and integrated sleep mode
- Simplified development



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Communications and Networking

- Bluetooth headsets
- ZigBee
- Integrated power management controllers
- Network storage link
- Wireless bridge
- Embedded clients

The Cortex-M3 processor delivers:

- Improved battery life
- Reduced memory requirements
- Simplified development and debug
- Rapid response to incoming data
- Outstanding data manipulation



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Spectrum of Applications

The Cortex-M3 processor addresses a huge range of application needs

Cost Sensitive	Low Power	High Performance
30+ DMIPS < \$1	25 DMIPS < 1mW ⁽¹⁾	120 DMIPS < 100MHz
32-bit MCUs at 8-bit prices - Minimal package < 20pins - Low-cost tools - Faster time to market	- Extended battery life - Integrated sleep modes - Rapid resume from sleep - System peripherals control	- Higher efficiency processor - Lower clock speeds - Greater determinism - Improved code density



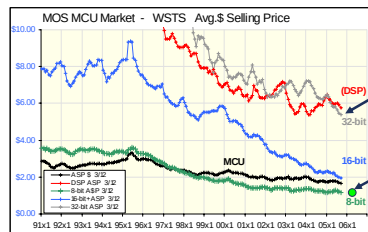
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Cortex-M3 Processor is Now Shipping

One of the Lead Partners for the Cortex-M3 processor introduced first \$1 ARM MCUs in early 2006, enabling ARM processors to range from \$1 to 1 GHz



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